

# MLOps Project Report

## LSTM Autoencoder Anomaly Detection

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## 1 Introduction

This report demonstrates an end-to-end MLOps pipeline for anomaly detection using an LSTM Autoencoder on the NASA CMAPSS FD001 dataset. The system integrates Airflow, MLflow, FastAPI, Flask, Prometheus, and Grafana.

## 2 Airflow (Data Ingestion)

Airflow is used to ingest sensor data periodically and store it in the database.

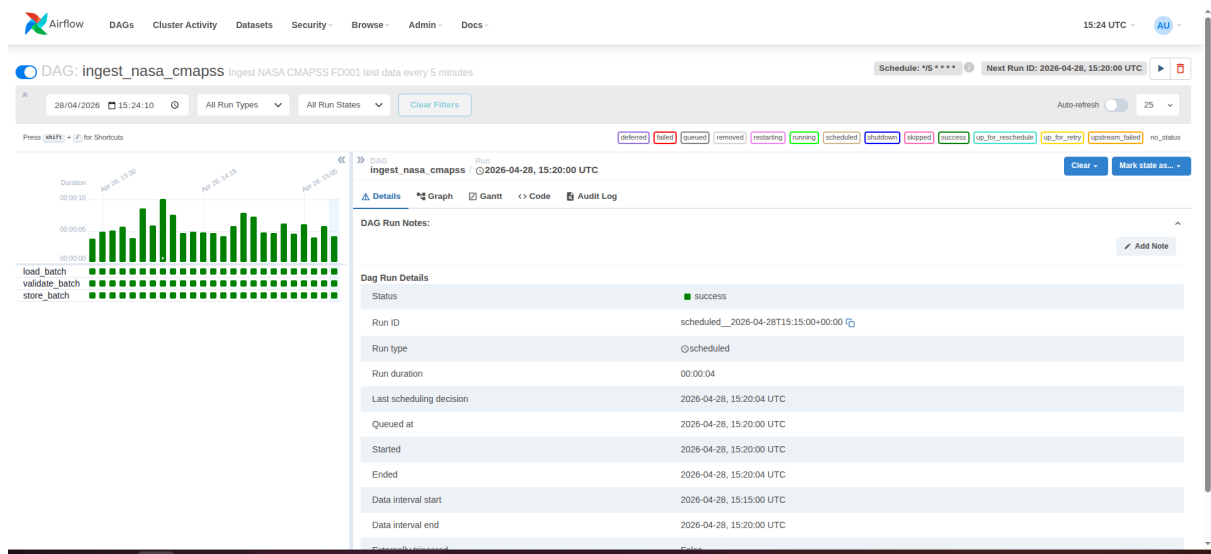


Figure 1: Airflow DAG Execution showing successful runs

### 3 Frontend Dashboard (Flask)

The Flask frontend displays real-time sensor values and anomaly predictions.

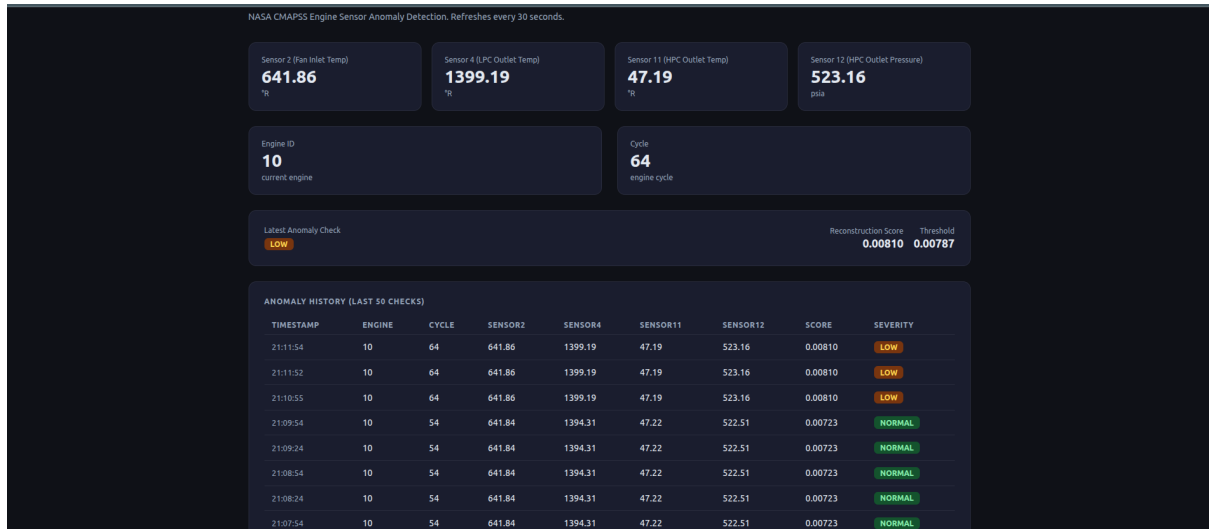


Figure 2: Frontend Dashboard showing live anomaly detection

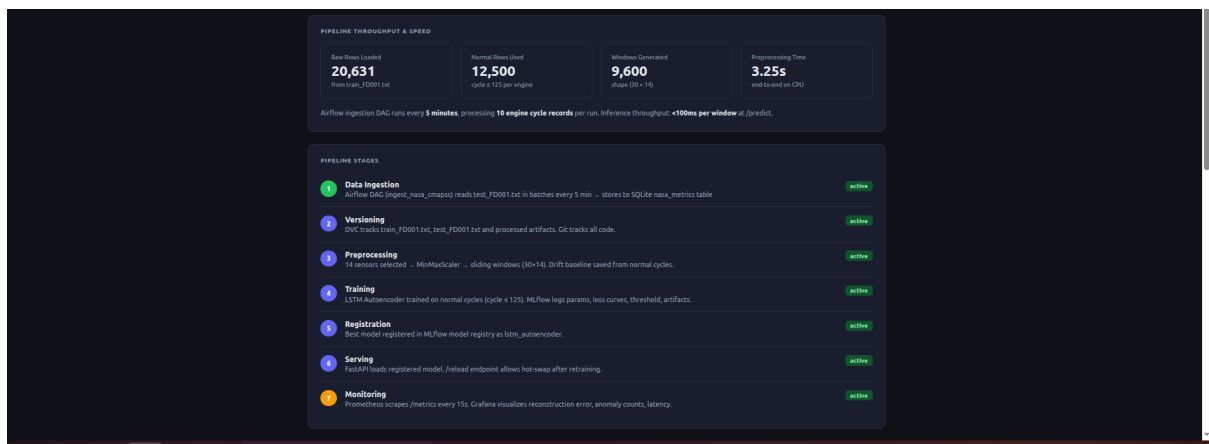


Figure 3: Pipeline Status and Throughput Visualization

## 4 MLflow (Experiment Tracking)

MLflow is used to track experiments, parameters, and metrics.

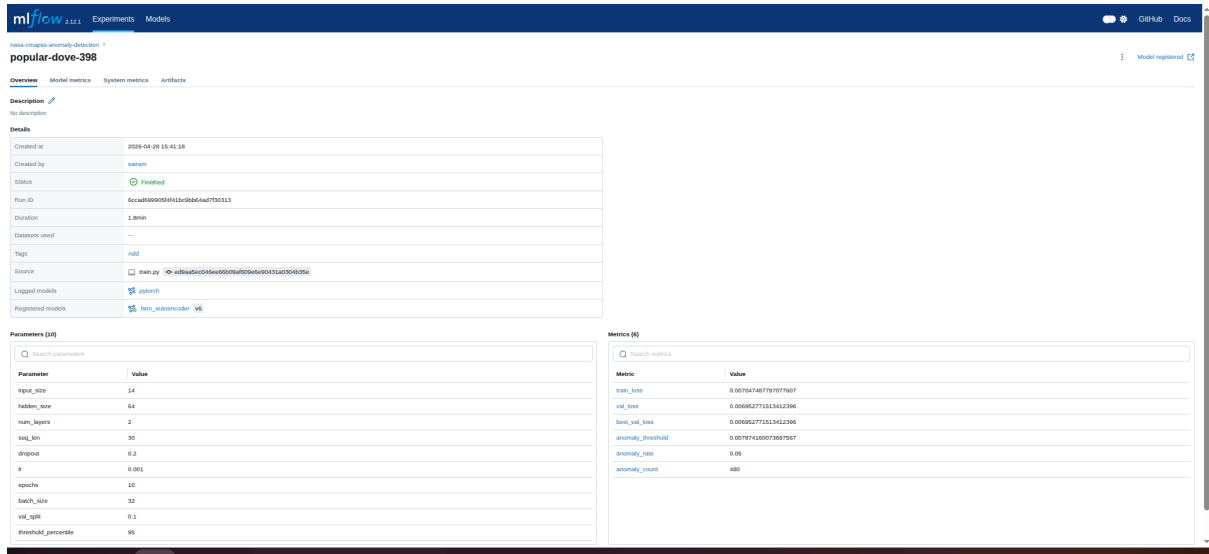


Figure 4: MLflow Experiment with parameters and metrics

## 5 Prometheus (Monitoring)

Prometheus collects metrics from the backend services.

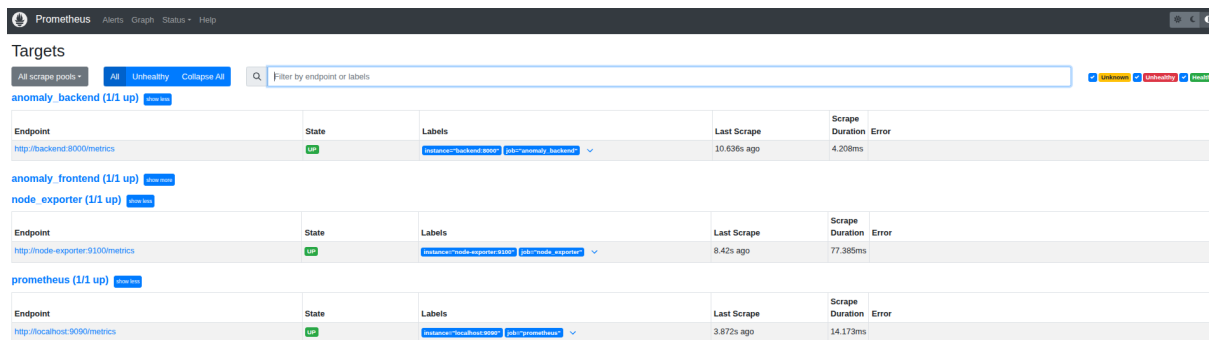


Figure 5: Prometheus Targets showing active services

## 6 Grafana (Visualization)

Grafana visualizes system and inference metrics using Prometheus data.

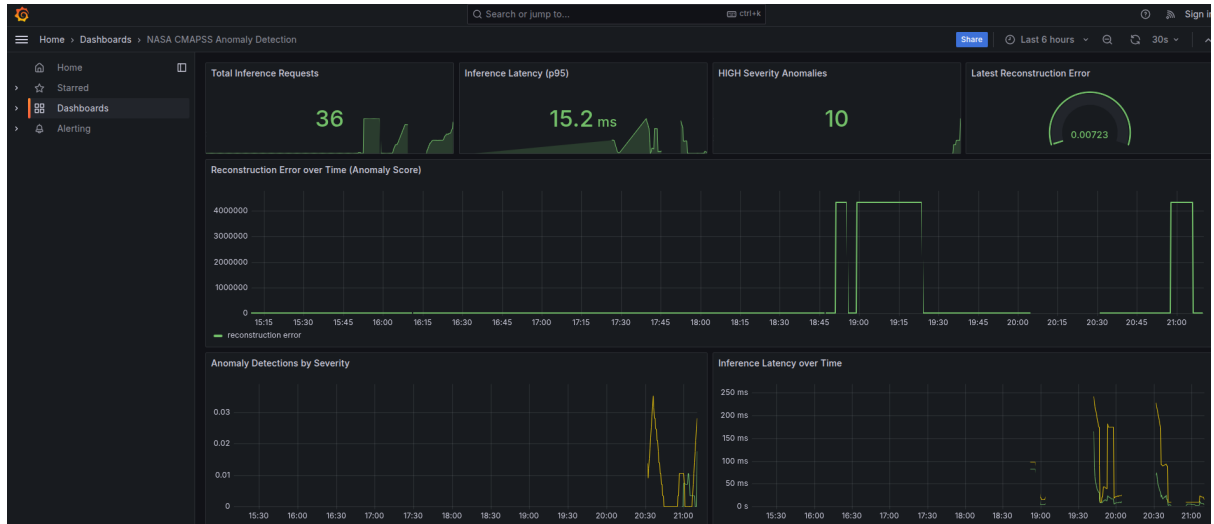


Figure 6: Grafana Dashboard showing inference metrics

## 7 Conclusion

The system successfully demonstrates a complete MLOps pipeline including data ingestion, model tracking, deployment, and monitoring. All components are integrated and functioning correctly.